

The Cosmic Microwave Background I

The expanding Universe, recombination, and acoustic oscillations

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Starting point and goal. No previous cosmology is assumed. We first build the homogeneous expanding Universe, then follow its cooling until neutral atoms form. Finally we add small perturbations and derive why the photon–baryon plasma oscillates. Lecture II will project those oscillations onto the sky, derive C_ℓ , and connect theory to real data.

Unless units are displayed explicitly, we use natural units $c = \hbar = k_B = 1$. We restore c occasionally when discussing distances or emphasizing their physical interpretation. This is the same convention used in the Particle Cosmology notes.

1 What are we trying to explain?

The cosmic microwave background (CMB) is electromagnetic radiation arriving from every direction. Its mean spectrum is extremely close to a blackbody,

$$B_\nu(T) = \frac{2h\nu^3}{c^2} \frac{1}{\exp(h\nu/k_B T) - 1}, \quad T_0 \simeq 2.725 \text{ K}. \quad (1)$$

The word *microwave* describes the wavelength today. The same photons had much higher energies in the past. The word *background* means that, after removing local emission, it fills the entire sky.

We write the temperature in direction $\hat{\mathbf{n}}$ as

$$T(\hat{\mathbf{n}}) = \bar{T} [1 + \Theta(\hat{\mathbf{n}})], \quad \Theta(\hat{\mathbf{n}}) \equiv \frac{\delta T(\hat{\mathbf{n}})}{\bar{T}}. \quad (2)$$

The sky is very uniform, but not perfectly so:

$$|\Theta| \sim 10^{-5}$$

after removal of the dipole induced mostly by our motion. These tiny variations are the central observable of these lectures.

Three facts that demand an explanation.

1. Why is the spectrum blackbody, and why is its temperature only 2.725 K today?
2. Why do photons reach us from a particular early epoch rather than from arbitrarily far back?
3. Why does the angular pattern contain a harmonic sequence of acoustic peaks?

The answers are, respectively: an expanding thermal Universe, recombination and decoupling, and sound waves in the primordial photon–baryon plasma.

The near-uniformity and the small variations carry complementary information. The mean temperature and spectrum probe the thermal history, while the 10^{-5} anisotropies probe gravity, matter content, geometry, and the initial conditions.

1.1 Inflation as the origin of the initial conditions

The near-uniformity itself raises a causal question. In a decelerating hot-Big-Bang history, many regions seen in widely separated directions on the last-scattering surface had no time to exchange signals before recombination. Nevertheless, their temperatures are almost identical. This is the horizon problem.

Inflation is a period of accelerated expansion,

$$\ddot{a} > 0, \quad \frac{d}{dt} \left(\frac{1}{aH} \right) < 0. \quad (3)$$

The second expression states that the comoving Hubble radius decreases. A region that was initially small enough to be causally connected can therefore be stretched to scales much larger than the Hubble radius. The observable Universe may then descend from one nearly homogeneous causal patch. After inflation ends, its energy is converted into a hot plasma and the familiar thermal evolution begins.

Inflation also supplies small deviations from homogeneity. Quantum fluctuations of the field driving inflation are stretched to macroscopic wavelengths. In the simplest single-field slow-roll picture,

$$\delta\phi_k \sim \frac{H}{2\pi}, \quad \mathcal{R}(\mathbf{k}) \simeq -\frac{H}{\dot{\phi}} \delta\phi_{\mathbf{k}}, \quad (4)$$

where $\mathcal{R}$ is the comoving curvature perturbation. Once a wavelength is larger than the Hubble radius, the growing adiabatic mode of $\mathcal{R}$ is approximately conserved. It later seeds the gravitational potentials and acoustic motion studied below.

A useful parametrization of its dimensionless power spectrum is

$$\Delta_{\mathcal{R}}^2(k) = A_s \left(\frac{k}{k_*} \right)^{n_s-1}. \quad (5)$$

Simple inflationary models predict perturbations that are nearly scale invariant (n_s close to one), nearly Gaussian, and predominantly adiabatic. They also select the same growing

solution for every Fourier mode. This synchronizes the subsequent acoustic phases and is essential for producing a sequence of sharp CMB peaks rather than a washed-out spectrum. Inflation can additionally generate tensor perturbations, whose amplitude relative to scalar perturbations is described by the tensor-to-scalar ratio r .

This short discussion provides initial conditions rather than a complete derivation of inflation. The background equations below explain what accelerated expansion means dynamically, while Lecture II shows how A_s , n_s , and r enter observable CMB statistics.

2 The homogeneous expanding Universe

2.1 Homogeneity and isotropy

On sufficiently large scales, the Universe has no preferred position and no preferred direction. These statements form the *cosmological principle*. They do not say that the Universe is exactly smooth on small scales: stars, galaxies, and clusters are the perturbations that we later place on top of the smooth background.

Homogeneity and isotropy constrain the spacetime metric to the Friedmann–Lemaître–Robertson–Walker (FLRW) form. For a spatially flat Universe,

$$\boxed{ds^2 = -dt^2 + a^2(t)\delta_{ij}dx^i dx^j.} \quad (6)$$

Here t is cosmic time, $\mathbf{x}$ is a *comoving* coordinate, and $a(t)$ is the scale factor. We set $a(t_0) = 1$ today. Two observers at fixed comoving coordinates have physical separation

$$d_{\text{phys}}(t) = a(t)|\Delta\mathbf{x}|. \quad (7)$$

Expansion means that $a(t)$ grows. It is space between freely moving comoving observers that changes; galaxies are not projectiles moving away from a central point.

2.2 The Hubble parameter and Hubble’s law

Differentiate the physical separation:

$$\dot{d}_{\text{phys}} = \frac{\dot{a}}{a}d_{\text{phys}} = H(t)d_{\text{phys}}, \quad H(t) \equiv \frac{\dot{a}}{a}. \quad (8)$$

In this subsection a dot means d/dt . For nearby objects this is Hubble’s law, $v = Hd$. The Hubble time H^{-1} is the characteristic expansion timescale; the Hubble length cH^{-1} is a useful characteristic distance. It is not, in general, the same as the age or the causal horizon.

2.3 Matter as a perfect fluid

The homogeneous contents are described by energy density $\rho(t)$ and pressure $P(t)$. For a perfect fluid,

$$T_{\mu\nu} = (\rho + P)U_\mu U_\nu + Pg_{\mu\nu}. \quad (9)$$

In the comoving frame $U^\mu = (1, 0, 0, 0)$. Einstein's equations applied to the flat FLRW metric give

$$\boxed{H^2 = \frac{8\pi G}{3}\rho,} \quad (10)$$

$$\frac{\ddot{a}}{a} = -\frac{4\pi G}{3}(\rho + 3P). \quad (11)$$

The first Friedmann equation says that the expansion rate is determined by the total energy density. The second says that pressure also gravitates.

It is often useful to introduce the reduced Planck mass,

$$M_{\text{Pl}}^{-2} = 8\pi G, \quad H^2 = \frac{\rho}{3M_{\text{Pl}}^2}.$$

2.4 Local energy conservation and the equation of state

In general relativity, energy and momentum obey the *local*, covariant conservation law $\nabla_\mu T^{\mu\nu} = 0$. For a homogeneous fluid this yields

$$\boxed{\dot{\rho} + 3H(\rho + P) = 0.} \quad (12)$$

This does *not* say that the total energy of an expanding Universe is constant. A general FLRW spacetime has no time-translation symmetry that would supply a globally conserved energy of the usual kind. In particular, the energy in a fixed comoving region need not be constant.

The local equation has a useful thermodynamic form. A region with fixed comoving volume has physical volume $V \propto a^3$ and fluid energy $E = \rho V$, so

$$d(\rho V) = -P dV.$$

This is the pressure-volume work relation: expansion changes the fluid energy when $P \neq 0$. For radiation, $E_\gamma \propto a^{-1}$ because photons redshift; for vacuum energy, $E_\Lambda \propto a^3$. Neither example violates local covariant conservation. The work analogy does not require an external reservoir into which the energy of redshifted photons must flow.

For a separately conserved component with $P = w\rho$ and constant w , Eq. (12) gives

$$\boxed{\rho(a) = \rho_0 a^{-3(1+w)}.} \quad (13)$$

Component	w	Evolution	Reason
Non-relativistic matter	0	$\rho_m \propto a^{-3}$	number dilution
Radiation	1/3	$\rho_r \propto a^{-4}$	dilution + redshift
Vacuum energy	-1	$\rho_\Lambda = \text{constant}$	energy of empty space

Inserting these scalings in the Friedmann equation gives

$$\begin{aligned} a(t) &\propto t^{1/2} && \text{during radiation domination,} \\ a(t) &\propto t^{2/3} && \text{during matter domination,} \\ a(t) &\propto e^{Ht} && \text{during vacuum domination.} \end{aligned} \tag{14}$$

Key result. The minimum background cosmology is $ds^2 = -dt^2 + a^2 d\mathbf{x}^2$, $H = \dot{a}/a$, $H^2 = 8\pi G\rho/3$, and $\dot{\rho} + 3H(\rho + P) = 0$. Everything in the CMB evolves on this background.

2.5 Redshift and the cooling of radiation

Consider two successive wave crests propagating through the expanding metric. Their physical separation stretches with a , so

$$\lambda \propto a, \quad \nu \propto a^{-1}, \quad E_\gamma = h\nu \propto a^{-1}. \tag{15}$$

Redshift is defined by

$$1 + z \equiv \frac{\lambda_{\text{obs}}}{\lambda_{\text{em}}} = \frac{a(t_0)}{a(t_{\text{em}})} = \frac{1}{a(t_{\text{em}})}. \tag{16}$$

A blackbody remains a blackbody under this uniform redshift, with

$$T(a) = \frac{T_0}{a} = T_0(1 + z). \tag{17}$$

This connects looking to large redshift with looking back to a hotter, denser Universe. At $z \simeq 1100$, Eq. (17) gives $T \simeq 3000$ K.

Radiation scales as a^{-4} rather than a^{-3} because there are two effects: the number of photons per physical volume gives a^{-3} , while the energy of every photon contributes one additional factor a^{-1} .

3 Light propagation, conformal time, and horizons

For photon propagation it is convenient to define conformal time

$$d\tau \equiv \frac{dt}{a(t)}. \tag{18}$$

The FLRW metric becomes

$$ds^2 = a^2(\tau) [-d\tau^2 + \delta_{ij} dx^i dx^j]. \tag{19}$$

Apart from the overall factor a^2 , this looks like Minkowski space. Introduce the *comoving radial coordinate* $\chi = |\mathbf{x}|$, with the observer at $\chi = 0$. At a fixed time the physical radial distance is $a(\tau)\chi$; in spherical coordinates $d\mathbf{x}^2 = d\chi^2 + \chi^2 d\Omega^2$. A radial light ray has $d\Omega = 0$ and $ds^2 = 0$, hence $|d\chi| = d\tau$ when $c = 1$. For a photon approaching us, $d\chi/d\tau = -1$: its

radial coordinate decreases as time increases. The positive comoving distance to an object observed at redshift z is consequently

$$\chi(z) = \tau_0 - \tau(z) = \int_0^z \frac{dz'}{H(z')}. \quad (20)$$

From here onward, as in the perturbation sections of the Particle Cosmology notes, a dot denotes a conformal-time derivative. We distinguish

$$\mathcal{H} \equiv \frac{\dot{a}}{a} = aH, \quad (21)$$

where H is the usual cosmic-time Hubble parameter.

The comoving particle horizon at time τ is the maximum comoving distance a signal could have travelled since the beginning,

$$d_H^{\text{com}}(\tau) = \tau - \tau_{\text{initial}} \simeq \tau. \quad (22)$$

Here the last equality chooses the origin of conformal time at the beginning. The comoving Hubble length is $(aH)^{-1} = \mathcal{H}^{-1}$. It measures an expansion timescale expressed as a distance, whereas the particle horizon counts the full distance light could have travelled. They are related but need not coincide. Pressure disturbances travel more slowly than light; their corresponding causal distance will be the *sound horizon*, introduced in Section 7.

3.1 Why the CMB is a two-dimensional image

If photons last scattered at conformal time τ_* , we observe them today at τ_0 on a sphere of comoving radius

$$\boxed{\chi_* = \tau_0 - \tau_*}. \quad (23)$$

A patch of comoving transverse size L_{com} at last scattering appears as an angular scale on this sphere. For a small angle,

$$\theta \simeq \frac{L_{\text{com}}}{\chi_*}. \quad (24)$$

Both lengths are comoving: using physical lengths instead multiplies numerator and denominator by the same a_* . This simple geometry will later connect sound waves in three dimensions to angular patterns on our sky.

4 A short thermal history of the photon bath

At sufficiently high temperature the Universe is a plasma. Photons, electrons, positrons, nuclei, neutrinos, and other species interact. A microscopic interaction rate Γ should be compared with the macroscopic expansion rate H :

$$\begin{aligned} \Gamma \gg H & : \text{ many interactions per expansion time; equilibrium,} \\ \Gamma \lesssim H & : \text{ interactions cannot keep up; decoupling.} \end{aligned} \quad (25)$$

This single comparison connects particle physics to cosmology.

For a species in thermal equilibrium, the distribution is

$$f(E) = \frac{1}{\exp[(E - \mu)/T] \pm 1}, \quad (26)$$

with $+$ for fermions and $-$ for bosons. Photons have vanishing chemical potential in full equilibrium. The photon number and energy densities scale as

$$n_\gamma = \frac{2\zeta(3)}{\pi^2} T^3, \quad \rho_\gamma = \frac{\pi^2}{15} T^4. \quad (27)$$

The observed blackbody spectrum therefore tells us that the radiation was once in excellent thermal equilibrium. Expansion subsequently redshifted the spectrum without changing its Planck form.

The broad sequence relevant here is

$$\begin{aligned} \text{hot ionized plasma} &\longrightarrow \text{neutral atoms form} \\ &\longrightarrow \text{photons free-stream} \longrightarrow \text{CMB observed today.} \end{aligned}$$

5 Recombination, decoupling, and last scattering

5.1 Why the early Universe was opaque

Free electrons Thomson-scatter photons with cross section

$$\sigma_T = \frac{8\pi}{3} \left(\frac{\alpha}{m_e} \right)^2 \quad (28)$$

in natural units. The scattering rate is roughly

$$\Gamma_T = n_e \sigma_T. \quad (29)$$

At early times n_e is large and $\Gamma_T/H \gg 1$. A photon travels only a short mean free path $\lambda_{\text{mfp}} = (n_e \sigma_T)^{-1}$ before scattering, so the plasma is opaque. Coulomb scattering couples electrons to protons; Thomson scattering couples electrons to photons. The three components behave approximately as one fluid.

5.2 Deriving the Saha equation: why recombination is below 13.6 eV

As the temperature falls, electrons and protons form neutral hydrogen,

$$p + e^- \leftrightarrow H^0 + \gamma.$$

First neglect helium and excited hydrogen levels. Here n_p counts *free protons*, n_e counts *free electrons*, and n_{H^0} counts neutral hydrogen atoms. A proton bound inside a neutral atom is included in n_{H^0} , not in the free-proton species density n_p . Thus

$$n_H = n_p + n_{H^0}$$

is the *total number density of hydrogen nuclei*, whether free or bound. Only in a fully ionized gas is $n_H \simeq n_p$. Recombination changes how nuclei are partitioned between free protons and neutral atoms; it does not destroy nuclei. Their number in a comoving volume is conserved, while their physical density n_H dilutes as a^{-3} .

Charge neutrality gives $n_e = n_p$. The useful ionization fraction is therefore

$$x_e \equiv \frac{n_e}{n_H}, \quad n_e = n_p = x_e n_H, \quad n_{H^0} = (1 - x_e) n_H.$$

Normalizing instead by the free-proton density would give $n_e/n_p = 1$ whenever free charges are present, and would not measure the fraction of hydrogen that is ionized.

From chemical equilibrium to number densities. For the reaction above, equilibrium requires

$$\mu_p + \mu_e = \mu_{H^0} + \mu_\gamma = \mu_{H^0},$$

because a blackbody photon bath has $\mu_\gamma = 0$. The atoms, protons, and electrons are non-relativistic and dilute. Integrating their Maxwell–Boltzmann distributions over momentum gives, with chemical potentials including rest energy,

$$n_i = g_i \int \frac{d^3p}{(2\pi)^3} e^{-(m_i + p^2/2m_i - \mu_i)/T} = g_i \left(\frac{m_i T}{2\pi} \right)^{3/2} e^{(\mu_i - m_i)/T}.$$

Substitute this expression for each of the three species before simplifying:

$$\begin{aligned} \frac{n_e n_p}{n_{H^0}} &= \frac{g_e g_p}{g_{H^0}} \left(\frac{m_e m_p}{m_{H^0}} \frac{T}{2\pi} \right)^{3/2} \\ &\times \exp\left(\frac{\mu_e + \mu_p - \mu_{H^0}}{T}\right) \exp\left(-\frac{m_e + m_p - m_{H^0}}{T}\right). \end{aligned}$$

Chemical equilibrium sets the first exponential to unity. The mass difference in the second is $m_e + m_p - m_{H^0} = E_I$, the hydrogen binding energy: the bound atom is lighter than its separated constituents. Using $m_{H^0} \simeq m_p$ *only in the prefactor* gives

$$\frac{n_e n_p}{n_{H^0}} \simeq \frac{g_e g_p}{g_{H^0}} \left(\frac{m_e T}{2\pi} \right)^{3/2} e^{-E_I/T}.$$

The small binding-energy mass difference must be retained in the exponential. For ground-state hydrogen, summing the spin states gives $g_e = 2$, $g_p = 2$, $g_{H^0} = 4$, so the degeneracy ratio is unity. Now explicitly substitute the free and neutral densities in terms of the ionization fraction:

$$\frac{n_e n_p}{n_{H^0}} = \frac{(x_e n_H)(x_e n_H)}{(1 - x_e) n_H} = n_H \frac{x_e^2}{1 - x_e}.$$

Equating the two expressions and dividing by n_H gives the Saha equation in this approximation,

$$\boxed{\frac{x_e^2}{1 - x_e} = \frac{1}{n_H} \left(\frac{m_e T}{2\pi} \right)^{3/2} e^{-E_I/T}, \quad E_I = 13.6 \text{ eV.}} \quad (30)$$

What sets the temperature? The guess $T \sim 13.6$ eV comes from comparing a typical thermal photon energy with the atomic ionization threshold. It is *not* a conclusion of the Saha equation. That equation also contains a large density-dependent prefactor. Define $\eta_H = n_H/n_\gamma$, of order 10^{-9} , and use the blackbody photon density:

$$\frac{x_e^2}{1-x_e} = \frac{\pi^2}{2\zeta(3)\eta_H} \left(\frac{m_e}{2\pi T}\right)^{3/2} e^{-E_I/T}.$$

At $T = E_I$ the right-hand side is enormous, implying $x_e \simeq 1$. To make it of order unity, the exponential must compensate the prefactor: one needs $E_I/T \sim 40$, not $E_I/T \sim 1$. For example, setting $x_e = 1/2$ and using a representative cosmological hydrogen abundance gives an equilibrium temperature near 0.32 eV. Physically, there are roughly a billion photons per baryon: even when a typical photon is too soft to ionize hydrogen, the high-energy tail can keep the gas ionized.

The word *recombination* refers to neutral atoms forming. The word *decoupling* refers to photons ceasing to scatter efficiently. The events are closely related but not identical. Saha determines an *equilibrium ionization fraction*, not the last-scattering time. Accurate calculations also require the departure from Saha equilibrium, excited atomic levels, the escape of Lyman- α photons, and two-photon decays. Last scattering occurs near $z_* \simeq 1100$, at $T_* \simeq 0.26$ eV (roughly 0.3 eV). We use Saha for the equilibrium reasoning; the scattering history below determines when photons can reach us.

5.3 Optical depth and the visibility function

Following the notation of the Particle Cosmology notes, define the optical depth from conformal time τ to today by

$$\tilde{\tau}(\tau) = \int_{\tau}^{\tau_0} a(\tau') n_e(\tau') \sigma_T d\tau'. \quad (31)$$

It follows from this definition that

$$\dot{\tilde{\tau}} = -an_e\sigma_T < 0, \quad -\dot{\tilde{\tau}} = an_e\sigma_T > 0. \quad (32)$$

This $\tilde{\tau}$ is not conformal time; the tilde is important. The probability that a photon travels from τ to us without scattering is $e^{-\tilde{\tau}}$. Therefore

$$\boxed{g(\tau) = -\dot{\tilde{\tau}} e^{-\tilde{\tau}}} \quad (33)$$

is the *visibility function*: the probability density that a photon observed today last scattered at time τ . It is sharply peaked at τ_* but has a finite width. Thus “the last-scattering surface” is a thin shell, not an infinitely thin mathematical sphere.

Quantity	Approximate value
Redshift of last scattering	$z_* \simeq 1100$
Temperature	$T_* \simeq 3000$ K $\simeq 0.3$ eV
Age of the Universe	$t_* \simeq 3.8 \times 10^5$ yr
Present mean CMB temperature	$T_0 \simeq 2.725$ K

Key result. Recombination removes free electrons, the Thomson mean free path grows rapidly, and the visibility function peaks. We see the CMB because the Universe changes from opaque to transparent.

6 From an exactly smooth Universe to anisotropies

An exactly FLRW Universe would produce an exactly uniform CMB. We therefore perturb the metric and matter fields around their homogeneous values,

$$\begin{aligned} g_{\mu\nu}(\tau, \mathbf{x}) &= \bar{g}_{\mu\nu}(\tau) + \delta g_{\mu\nu}(\tau, \mathbf{x}), \\ \rho_i(\tau, \mathbf{x}) &= \bar{\rho}_i(\tau)[1 + \delta_i(\tau, \mathbf{x})]. \end{aligned} \quad (34)$$

The observed amplitude 10^{-5} motivates linear perturbation theory: keep only terms first order in δ_i , metric potentials, velocities, and Θ .

Scalar, vector, and tensor perturbations evolve independently at linear order. Density perturbations are scalar, so we focus on scalar modes. In conformal Newtonian gauge, using precisely the convention of the Particle Cosmology notes,

$$ds^2 = a^2(\tau) \left[-(1 + 2\Psi)d\tau^2 + (1 - 2\Phi)\delta_{ij}dx^i dx^j \right]. \quad (35)$$

Ψ acts like a Newtonian gravitational potential and Φ is the perturbation of spatial curvature. Equation (35) is a *perturbed metric*; the perturbed Einstein equations determine how these potentials respond to matter.

Why just two scalar potentials? Before choosing coordinates, the scalar part of a metric perturbation contains four functions: a time–time perturbation, a scalar time–space shift, a spatial trace, and a scalar spatial shear. There are two scalar coordinate freedoms: a change of time slicing and a longitudinal change of spatial coordinates. Newtonian gauge uses them to set the shift and scalar shear to zero, leaving Ψ and Φ . This counts scalar metric functions after a coordinate choice, *not two independently propagating gravitational degrees of freedom*. The scalar Einstein constraint equations relate the potentials to density and momentum perturbations. Matter supports scalar dynamics, including the sound waves we derive below. The two freely propagating gravitational degrees of freedom of general relativity are the two *tensor* polarizations of gravitational waves, outside our present scalar discussion.

What is anisotropic stress, and why do neutrinos produce it? Pressure is momentum flux: particles crossing a surface carry momentum through it. An isotropic distribution has the same pressure in every direction. Its departure from isotropy is the traceless part of the spatial stress tensor,

$$T^i_j = P\delta^i_j + \pi^i_j, \quad \pi^i_i = 0.$$

The tensor π^i_j is called *anisotropic stress*. In particle language it comes from a quadrupole in the angular distribution of momenta, after removing the isotropic pressure and bulk motion. A scalar perturbation can produce such a quadrupole: its spatial gradient supplies a preferred direction locally, without making the background Universe anisotropic.

After neutrinos decouple, they travel freely from regions with different densities and temperatures. At a given point, particles arriving from different directions therefore have different distributions. With few collisions to isotropize those distributions, a quadrupole develops. Tightly coupled photons instead scatter often, which suppresses their quadrupole; non-relativistic baryons and cold dark matter have very small random-velocity stresses. Neutrinos are not unique in principle: photons also develop anisotropic stress as tight coupling fails and after decoupling. The traceless spatial Einstein equation relates this stress to $\Phi - \Psi$. For negligible total scalar anisotropic stress, $\Phi = \Psi$; free-streaming neutrinos make that equality approximate in the real Universe.

6.1 Fourier modes simplify linear evolution

A sound wave is a spatial pattern of alternating overdensities and underdensities. We decompose an arbitrary small perturbation into sinusoidal patterns, or Fourier modes. The vector $\mathbf{k}$ specifies the direction of variation and its magnitude $k = |\mathbf{k}|$ the inverse length scale. Write, for example,

$$\delta_\gamma(\tau, \mathbf{x}) = \int \frac{d^3k}{(2\pi)^3} \delta_\gamma(\tau, \mathbf{k}) e^{i\mathbf{k}\cdot\mathbf{x}}. \quad (36)$$

Because the background is homogeneous and the equations are linear, different $\mathbf{k}$ modes evolve independently. A spatial derivative becomes multiplication by $i\mathbf{k}$; coupled partial differential equations become ordinary differential equations in time for each k .

The comoving wavelength is $\lambda_{\text{com}} = 2\pi/k$; k^{-1} is the scale over which the phase changes by one radian, not a full wavelength. Comparing the gradient scale with the comoving Hubble length, we call a mode super-Hubble when $k \ll \mathcal{H}$ and sub-Hubble when $k \gg \mathcal{H}$. For acoustic evolution the more precise comparison is with the sound horizon: appreciable phase evolution requires $kr_s \gtrsim 1$. At last scattering, a transverse scale of order k^{-1} subtends an angle of order $1/(k\chi_*)$. Lecture II will describe angular patterns by spherical-harmonic multipoles ℓ , with the approximate correspondence $\ell \sim k\chi_*$.

For photons $\rho_\gamma \propto T^4$. Linearizing gives

$$\frac{\delta\rho_\gamma}{\bar{\rho}_\gamma} = 4\frac{\delta T}{\bar{T}}, \quad \boxed{\Theta_0 = \frac{1}{4}\delta_\gamma}. \quad (37)$$

Here Θ_0 is the angle-averaged photon-temperature perturbation for a Fourier mode. It should not be confused with today's mean temperature T_0 .

More precisely, at each $(\tau, \mathbf{x})$ the photon perturbation also depends on the propagation direction $\hat{\mathbf{n}}$. For a scalar Fourier mode define $\mu = \hat{\mathbf{k}} \cdot \hat{\mathbf{n}}$ and expand the angular dependence as

$$\Theta(\tau, \mathbf{k}, \mu) = \sum_{\ell=0}^{\infty} (-i)^\ell (2\ell+1) \Theta_\ell(\tau, k) P_\ell(\mu). \quad (38)$$

The monopole Θ_0 is the directional average and determines the photon energy density. The dipole Θ_1 is the forward-backward asymmetry and determines the net photon flux, or momentum density. It is therefore often called the *photon temperature dipole*, the *photon*

dipole, or, more loosely, the *photon velocity*. With this multipole convention and the Fourier convention used here,

$$\boxed{\theta_\gamma = 3k\Theta_1.} \quad (39)$$

The factor and sign are convention dependent. We reserve Θ_1 for the angular dipole, v_γ for a scalar velocity amplitude, and θ_γ for the velocity divergence. These are local moments of the photon momentum directions; they are conceptually distinct from the observed whole-sky coefficients $a_{\ell m}$.

6.2 Where did the perturbations come from?

Here we take the initial conditions as given: an early mechanism such as inflation produces an almost scale-invariant, nearly Gaussian, predominantly adiabatic curvature perturbation $\mathcal{R}_i(\mathbf{k})$. “Adiabatic” means that all species initially describe the same local time shift; there is a single growing scalar mode. On scales larger than the Hubble radius, $\mathcal{R}$ is approximately conserved. Lecture II will use its power spectrum $\Delta_{\mathcal{R}}^2(k)$, in the notation of the course notes.

Fourier modes make the three-dimensional evolution simple, even though the CMB is observed on a sphere. Spherical harmonics provide the corresponding natural basis for the two-dimensional observation. Lecture II connects the two descriptions.

7 The tightly coupled photon–baryon fluid

7.1 The physical picture

Before recombination photons scatter so rapidly that their mean free path is much smaller than the wavelength of interest. Photons and baryons share a common bulk velocity,

$$\mathbf{v}_\gamma \simeq \mathbf{v}_b. \quad (40)$$

Gravity tries to compress an overdense region. Photon pressure resists compression. The result is the cosmological analogue of sound in air, except that photons provide most of the pressure while photons and baryons both provide inertia.

The inertia of a relativistic fluid is set by its *enthalpy density*, $\rho + P$: to first order in velocity its momentum density is $(\rho + P)\mathbf{v}$ in a local orthonormal frame. Photons therefore contribute $4\bar{\rho}_\gamma/3$ to the inertia, whereas non-relativistic baryons contribute approximately $\bar{\rho}_b$. Define their ratio, the baryon-loading ratio used in the course notes,

$$\boxed{R \equiv \frac{3\bar{\rho}_b}{4\bar{\rho}_\gamma}.} \quad (41)$$

Thus *baryon inertia* is the physical resistance of baryons to acceleration, while *baryon loading* describes how much of it is added relative to the photon inertia. The total inertia is $(4\bar{\rho}_\gamma/3)(1 + R)$. Since $\rho_b \propto a^{-3}$ and $\rho_\gamma \propto a^{-4}$, $R \propto a$: the baryon contribution becomes more important *relative to photons*, even though both densities decrease. In conformal time, $\dot{R} = \mathcal{H}R$.

For an adiabatic compression, the baryon-to-photon number ratio stays fixed. Since $n_\gamma \propto T^3$ and $\rho_\gamma \propto T^4$, this gives $\delta_b = 3\delta_\gamma/4$, or $\delta\rho_b = R\delta\rho_\gamma$. Using $\delta P_b \simeq 0$ and $\delta P_\gamma = \delta\rho_\gamma/3$, the sound speed is

$$c_s^2 \equiv \frac{\delta P}{\delta\rho} = \frac{\delta P_\gamma}{\delta\rho_\gamma + \delta\rho_b} = \boxed{\frac{1}{3(1+R)}}. \quad (42)$$

With no baryons, $c_s = 1/\sqrt{3}$. Adding baryons lowers the sound speed because they add inertia but essentially no pressure.

7.2 Continuity and Euler equations

We need two statements: how compression changes a density, and how pressure, gravity, and collisions change a velocity. These are the energy and momentum moments of the photon Boltzmann equation, or equivalently the continuity and Euler equations of the fluid. Take one Fourier mode and define its velocity divergence $\theta_i \equiv i\mathbf{k} \cdot \mathbf{v}_i$. Lowercase θ_i is a velocity-divergence variable; uppercase Θ_0 is a temperature perturbation. In real space, negative divergence means converging flow and hence compression. A spatially uniform velocity has zero divergence, so bulk translation by itself does not compress the fluid.

Continuity: compression changes photon temperature. In conformal Newtonian gauge the photon continuity equation is

$$\dot{\delta}_\gamma = -\frac{4}{3}\theta_\gamma + 4\dot{\Phi}, \quad (43)$$

The factor $4/3 = (\bar{\rho}_\gamma + \bar{P}_\gamma)/\bar{\rho}_\gamma$ accounts for both energy transport and pressure work. A useful way to see the metric term is that a small coordinate cell has proper volume proportional to $a^3(1 - 3\Phi)$. A changing Φ changes the local volume expansion; the effective perturbation of its expansion rate is $\theta_\gamma - 3\dot{\Phi}$. Thus the continuity equation can be read as $\dot{\delta}_\gamma = -(4/3)(\theta_\gamma - 3\dot{\Phi})$. Since $\Theta_0 = \delta_\gamma/4$, it also says

$$\dot{\Theta}_0 = -\frac{\theta_\gamma}{3} + \dot{\Phi}.$$

Compression raises the photon temperature, while changing spatial curvature also changes the photon energy density measured locally.

Euler: what accelerates each component? The photon Euler equation, neglecting its quadrupole in tight coupling, is

$$\dot{\theta}_\gamma = k^2 \left(\frac{1}{4}\delta_\gamma + \Psi \right) + (-\dot{\tau})(\theta_b - \theta_\gamma). \quad (44)$$

The pressure acceleration is $-\nabla\delta P_\gamma/(\bar{\rho}_\gamma + \bar{P}_\gamma) = -\nabla\Theta_0$; the gravitational acceleration is $-\nabla\Psi$. Taking their divergences gives $k^2\Theta_0$ and $k^2\Psi$, since $\nabla^2 \rightarrow -k^2$. The remaining term is Thomson drag. Its positive rate $-\dot{\tau} = an_e\sigma_T$ is the number of scatterings per unit *conformal* time.

The baryon Euler equation, neglecting baryon thermal pressure, is

$$\dot{\theta}_b + \mathcal{H}\theta_b = k^2\Psi + \frac{-\dot{\tau}}{R}(\theta_\gamma - \theta_b). \quad (45)$$

Here $\mathcal{H}\theta_b$ is the expansion drag: in the absence of forces a non-relativistic peculiar velocity decreases as a^{-1} . There is no corresponding term for the photon bulk-velocity variable, whose radiation pressure and background redshifting cancel it. Baryons feel gravity but contribute negligible pressure.

Why scattering equalizes velocities. If photons have a bulk drift relative to the electrons, their radiation is dipolar in the electron rest frame. Scattering transfers momentum from this radiation flux to electrons, and Coulomb interactions share it with the ions. Thus photons are dragged toward the baryon velocity, and baryons toward the photon velocity. The two *force densities* are equal and opposite. Dividing them by the two inertias gives different accelerations: the baryon acceleration contains $(4\bar{\rho}_\gamma/3)/\bar{\rho}_b = 1/R$. Indeed, the collision contribution to total momentum obeys

$$\frac{4\bar{\rho}_\gamma}{3}\dot{\theta}_\gamma|_{\text{coll}} + \bar{\rho}_b\dot{\theta}_b|_{\text{coll}} = 0.$$

Opposite signs *and this inertia weighting* are both necessary for momentum conservation.

To check relaxation explicitly, define the velocity slip $\Delta\theta = \theta_\gamma - \theta_b$. Retaining only the collision terms gives

$$\left.\frac{d\Delta\theta}{d\tau}\right|_{\text{coll}} = -(-\dot{\tau})\left(1 + \frac{1}{R}\right)\Delta\theta.$$

The coefficient on the right is negative: either sign of slip decays exponentially when the rate is approximately constant. Pressure and expansion continually source a small slip, but rapid scattering keeps it small. Tight coupling requires the scattering rate to exceed both the expansion and mode-evolution rates (in particular $-\dot{\tau} \gg \mathcal{H}, k$ is sufficient). It locks *bulk velocities*, not the density contrasts or the temperature perturbations of the two species.

The original Particle Cosmology notes instead write the scalar velocity as

$$\mathbf{v}_i = i\hat{\mathbf{k}} v_i. \quad (46)$$

With the Fourier convention used here, the translation is

$$\boxed{\theta_i = i\mathbf{k} \cdot \mathbf{v}_i = -kv_i.} \quad (47)$$

Thus v_i in the long notes and θ_i here contain the same scalar velocity information; the extra factor and sign come only from the definition.

7.3 Deriving the acoustic oscillator

First derive the equation for the actual photon temperature. Add Eq. (44) to R times Eq. (45). The collision terms cancel. This order matters: a small velocity slip multiplied by a large scattering rate can transfer a finite momentum. Setting the collision

terms to zero in each species equation separately would lose that transfer. Only after adding the equations take the leading tight-coupling limit $\theta_\gamma \simeq \theta_b \equiv \theta$ to obtain

$$(1 + R)\dot{\theta} + \mathcal{H}R\theta = k^2\Theta_0 + (1 + R)k^2\Psi.$$

Gravity accelerates the whole fluid; only photons supply the pressure force. Dividing by $1 + R$ makes the reduced pressure acceleration explicit:

$$\dot{\theta} + \frac{\dot{R}}{1 + R}\theta = \frac{k^2}{1 + R}\Theta_0 + k^2\Psi, \quad \dot{R} = \mathcal{H}R.$$

Now differentiate $\dot{\Theta}_0 = -\theta/3 + \dot{\Phi}$, substitute the combined Euler equation, and use $\theta = -3(\dot{\Theta}_0 - \dot{\Phi})$. This gives

$$\ddot{\Theta}_0 + \frac{\dot{R}}{1 + R}\dot{\Theta}_0 + k^2c_s^2\Theta_0 = \ddot{\Phi} + \frac{\dot{R}}{1 + R}\dot{\Phi} - \frac{k^2}{3}\Psi.$$

This equation is already a complete temperature oscillator at leading order in tight coupling, for specified gravitational potentials. The pressure term restores the fluid, $\dot{R}/(1 + R)$ accounts for the changing inertia, $-k^2\Psi/3$ is gravitational compression, and the derivatives of Φ arise from the changing spatial metric in the continuity equation.

Why introduce an effective temperature? The intrinsic perturbation Θ_0 tells us how hot the plasma is *where the photon is emitted*. The photon also undergoes a gravitational redshift on its way out. For time-independent potentials, after factoring out the background redshift, the fractional energy change between emission and observation is $\Psi_* - \Psi_{\text{obs}}$. The observer term is common to all directions and contributes only a monopole. The emission term therefore adds Ψ_* to the intrinsic temperature perturbation. This motivates

$$X \equiv \Theta_0 + \Psi. \quad (48)$$

In our sign convention a potential well has $\Psi < 0$, so its redshift reduces the temperature received from an intrinsically hot region. Thus X is the intrinsic temperature *plus the local gravitational contribution*, not a new physical degree of freedom. It is the emission contribution used in Lecture II; Doppler shifts and evolving potentials along the path add further terms to the observable. We could keep using Θ_0 , but X makes both the observed combination and the gravity-shifted oscillation particularly transparent.

Substitute $\Theta_0 = X - \Psi$ into the temperature equation:

$$\ddot{X} + \frac{\dot{R}}{1 + R}\dot{X} + k^2c_s^2X = \ddot{\Phi} + \ddot{\Psi} + \frac{\dot{R}}{1 + R}(\dot{\Phi} + \dot{\Psi}) - k^2c_s^2R\Psi. \quad (49)$$

This is a damped, driven harmonic oscillator for every Fourier mode k :

- kc_s is the acoustic frequency in conformal time; the restoring force originates in photon pressure;
- $\dot{R}/(1 + R)$ changes the amplitude as the baryon inertia grows. This is expansion-related damping, not dissipative photon diffusion;
- $-k^2c_s^2R\Psi$ shifts the equilibrium even for constant potentials;
- time derivatives of Φ and Ψ drive the oscillation when gravity evolves.

A solvable limit and the acoustic phase. For intuition, freeze R , Φ , and Ψ over an oscillation, so their time derivatives vanish. Equation (49) reduces to

$$\ddot{X} + k^2 c_s^2 (X + R\Psi) = 0.$$

The displacement from equilibrium is therefore $X + R\Psi$, whose solution is

$$X(\tau) + R\Psi = C_1 \cos[kr_s(\tau)] + C_2 \sin[kr_s(\tau)], \quad (50)$$

where

$$r_s(\tau) = \int_0^\tau c_s(\tau') d\tau' \quad (51)$$

is the comoving sound horizon: the maximum comoving distance a sound wave could have travelled by time τ . For the constant-coefficient toy model, $r_s = c_s \tau$ (with the chosen time origin), and Eq. (50) is exact. When R varies slowly, the accumulated phase is still approximately $k \int c_s d\tau = kr_s$, but the amplitude and the equilibrium also evolve: constant C_1, C_2 are then only a toy approximation. Rapid potential evolution requires solving the driven equation.

For the dominant adiabatic growing mode, all wavelengths begin with a coherent phase: they are initially displaced with negligible velocity. Consequently the cosine term dominates. At recombination we take a snapshot. Modes satisfying

$$k_n r_s(\tau_*) \simeq n\pi, \quad n = 1, 2, 3, \dots \quad (52)$$

are at extrema of compression or rarefaction. This harmonic series in k becomes the series of acoustic peaks in angular multipole ℓ after projection onto the sky.

Key result. Gravity displaces the photon–baryon fluid, photon pressure restores it, inflationary/adiabatic initial conditions synchronize the phase, and recombination takes a snapshot. These four facts produce the acoustic peaks.

7.4 Baryon loading and gravitational driving

We can now derive the equilibrium shift rather than assume it. For a static potential and fixed R , set the velocity and acceleration to zero in the combined Euler equation. Pressure must balance gravity:

$$0 = \frac{k^2}{1+R} \Theta_{0,\text{eq}} + k^2 \Psi, \quad \Theta_{0,\text{eq}} = -(1+R)\Psi.$$

Photons must provide enough pressure to support *both* their own inertia and the baryons in the potential well. More baryons therefore require a larger intrinsic temperature and density perturbation for balance. Converting to the effective temperature gives

$$X_{\text{eq}} = \Theta_{0,\text{eq}} + \Psi = -R\Psi.$$

With no baryons the intrinsic equilibrium temperature $-\Psi$ is exactly cancelled by the gravitational redshift, so $X_{\text{eq}} = 0$. Baryons leave the extra offset $-R\Psi$. For a potential well

($\Psi < 0$), this offset is positive. An oscillation of amplitude A has extrema $X = -R\Psi \pm A$: its compressed, hot phase is enhanced relative to its rarefied, cold phase. Projecting these coherent oscillations gives the characteristic odd–even temperature-peak modulation: odd compression peaks are enhanced as the baryon density increases.

During exact matter domination, the gravitational potentials are nearly constant. Around radiation–matter equality they evolve. A time-dependent equilibrium point does work on the oscillator. Modes entering the horizon during radiation domination are therefore boosted by *radiation driving*. This is why the peak envelope also contains information about the physical matter density and the time of equality.

The acoustic peaks are evidence for coherent initial conditions. If different regions or Fourier modes began oscillating with random phases, averaging over them would wash out the peak–trough sequence.

8 Why the peaks do not continue forever

Tight coupling is an approximation. Photons have a small but nonzero mean free path and random-walk out of hot overdense regions. Heat conduction and shear viscosity erase anisotropy below a diffusion length. In Fourier space the acoustic amplitude is approximately multiplied by

$$\exp[-k^2/k_D^2]. \quad (53)$$

This is Silk, or photon-diffusion, damping. The finite thickness of the visibility function adds further averaging: photons last-scattering at slightly different times sample slightly different oscillation phases.

The main physical scales at last scattering are therefore

- the comoving distance χ_* from us to the last-scattering surface;
- the sound horizon $r_s(\tau_*)$, setting the acoustic spacing;
- the diffusion length $r_D \sim k_D^{-1}$, setting the damping tail;
- the horizon around equality, setting gravitational driving.

Projection turns the sound horizon into an angle,

$$\boxed{\theta_s = \frac{r_s(\tau_*)}{\chi_*}, \quad \ell_A \simeq \frac{\pi}{\theta_s}.} \quad (54)$$

This compact equation is why the CMB constrains geometry: early-time physics predicts the ruler r_s , while late-time expansion determines the distance χ_* .

Summary

We can now tell the CMB story without equations:

1. An FLRW Universe expands and cools; photon wavelengths grow as a and $T \propto a^{-1}$.
2. The early Universe is an opaque photon–baryon plasma maintained by Thomson scattering.
3. Neutral hydrogen formation rapidly lowers n_e ; the visibility function peaks and photons free-stream.
4. Small primordial perturbations provide gravitational potential wells.
5. Gravity and photon pressure make the tightly coupled plasma oscillate with sound speed $c_s^2 = 1/[3(1 + R)]$.
6. Recombination photographs these coherently phased oscillations. Baryons modulate compressions and diffusion damps short wavelengths.
7. We observe the three-dimensional pattern projected onto a two-dimensional sphere. Lecture II turns this statement into the angular spectrum C_ℓ .

Companion numerical notebook

The physical sequence in this lecture can be explored in the accompanying notebook:

[Open the Lecture I exercise directly in Google Colab.](#)

It visualizes radiation, matter, and vacuum domination; solves the Saha equation; constructs an approximate visibility function; calculates the photon–baryon sound horizon; and plots the coherent density and velocity phases. The notebook emphasizes where the Saha approximation differs from a precision recombination calculation and ends with exercises exploring the physical parameters.

Further derivation A: scale-factor solutions

For a component with constant w , combine $\rho \propto a^{-3(1+w)}$ with Friedmann:

$$\frac{1}{a} \frac{da}{dt} \propto a^{-3(1+w)/2}.$$

For $w \neq -1$, separation of variables gives

$$a(t) \propto t^{\frac{2}{3(1+w)}}. \tag{55}$$

Thus radiation gives $a \propto t^{1/2}$ and matter gives $a \propto t^{2/3}$.

Further derivation B: why the large-scale temperature is gravitational

The photon reaching us carries an intrinsic temperature perturbation and is redshifted while climbing out of a potential well. At last scattering these combine as

$$\left[\frac{1}{4} \delta_\gamma + \Psi \right]_* . \quad (56)$$

For an adiabatic mode on scales much larger than the sound horizon during matter domination, the result is the ordinary Sachs–Wolfe relation

$$\frac{\delta T}{T} \Big|_{\text{sw}} \simeq \frac{1}{3} \Psi_* . \quad (57)$$

There is also a Doppler contribution from the baryon velocity and an integrated Sachs–Wolfe contribution if Φ or Ψ evolves along the photon path. These are the opening ingredients of Lecture II.

Notation guide

Symbol	Meaning
$t, H = a^{-1} da/dt$	cosmic time and cosmic-time Hubble rate; dot means d/dt in Sec. 2
$\tau, \mathcal{H} = \dot{a}/a = aH$	conformal time and conformal Hubble rate; thereafter dot means $d/d\tau$
$\tilde{\tau}$	Thomson optical depth, not a time coordinate
χ, χ_*	comoving radial coordinate and distance to last scattering
n_H, n_{H^0}, x_e	total hydrogen-nucleus density, neutral-atom density, and n_e/n_H
Φ, Ψ	conformal-Newtonian-gauge metric potentials
$\Theta = \delta T/\bar{T}$	fractional photon-temperature perturbation
δ_γ	fractional photon-energy-density perturbation; $\Theta_0 = \delta_\gamma/4$
Θ_1	photon temperature dipole; proportional to photon momentum or bulk flow
$\theta_i = i\mathbf{k} \cdot \mathbf{v}_i$	velocity divergence, distinct from temperature Θ_0
$X = \Theta_0 + \Psi$	intrinsic temperature plus local gravitational contribution
$R = 3\bar{\rho}_b/(4\bar{\rho}_\gamma)$	baryon-to-photon inertia ratio (baryon loading)
$\mathcal{R}, \Delta_{\mathcal{R}}^2$	comoving curvature perturbation and its dimensionless power spectrum
r_s	comoving sound horizon

Source and scope note

This lecture is a shortened, reordered version of the Particle Cosmology course notes: FLRW dynamics and redshift from Lectures 1–2; equilibrium, decoupling, and recombination from

Lecture 3; scalar perturbations and conformal Newtonian gauge from Lecture 4; and tight coupling and CMB acoustic physics from Lecture 5. The notation has been aligned with those notes. Material has been reordered so that no prior cosmology course is assumed; detailed gauge transformations and the full Boltzmann hierarchy are postponed. The intermediate steps on chemical equilibrium, momentum exchange, and the oscillator's equilibrium have been expanded here for a first introduction. For supplementary derivations see [Ma & Bertschinger \(1995\)](#) for the fluid equations and [Hu & Dodelson's discussion of baryon loading](#) for the oscillator interpretation. Metric signs throughout follow the Particle Cosmology master notes.